

$$C_{dB\phi_1}^{i1-i2-} \rightarrow$$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(\frac{1}{216} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0} [m_1, m_d^{-i2}, m_q^{-i1}] + \frac{1}{144} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,2,1,-1} [m_1, m_d^{-i2}, m_q^{-i1}] - \right. \\ & \frac{1}{216} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1} [m_1, m_d^{-i2}, m_q^{-i1}] + \frac{1}{18} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0} [m_1, m_d^{-i2}, \tilde{\mu}] - \\ & \frac{1}{18} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1} [m_1, m_d^{-i2}, \tilde{\mu}] - \frac{1}{144} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,2,1,-1} [m_1, m_q^{-i1}, m_d^{-i2}] - \\ & \frac{1}{72} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0} [m_1, m_q^{-i1}, \tilde{\mu}] + \frac{1}{72} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1} [m_1, m_q^{-i1}, \tilde{\mu}] - \\ & \frac{1}{72} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_d^{-i2}] - \frac{1}{144} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^{-i1}] - \\ & \frac{1}{8} g_1 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} LF_{2,1,1,0} [m_2, m_q^{-i1}, \tilde{\mu}] + \frac{1}{8} g_1 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} LF_{3,1,1,-1} [m_2, m_q^{-i1}, \tilde{\mu}] - \\ & \frac{1}{16} g_1 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} LF_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^{-i1}] - \frac{1}{9} g_1 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} LF_{2,1,1,0} [m_3, m_d^{-i2}, m_q^{-i1}] - \\ & \frac{1}{6} g_1 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} LF_{2,2,1,-1} [m_3, m_d^{-i2}, m_q^{-i1}] + \frac{1}{9} g_1 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} LF_{3,1,1,-1} [m_3, m_d^{-i2}, m_q^{-i1}] + \\ & \frac{1}{6} g_1 m_3 \tilde{\mu} g_3^2 y_d^{i1i2} LF_{2,2,1,-1} [m_3, m_q^{-i1}, m_d^{-i2}] - \frac{1}{16} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} LF_{2,2,1,-1} [m_q^p, \tilde{\mu}, m_u^r] + \\ & \frac{1}{16} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} LF_{2,2,1,-1} [m_u^r, \tilde{\mu}, m_q^p] + \\ & \frac{1}{18} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0} [\tilde{\mu}, m_1, m_d^{-i2}] + \frac{1}{36} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1} [\tilde{\mu}, m_1, m_d^{-i2}] - \\ & \frac{1}{72} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{2,1,1,0} [\tilde{\mu}, m_1, m_q^{-i1}] - \frac{1}{36} m_1 \tilde{\mu} g_1^3 y_d^{i1i2} LF_{3,1,1,-1} [\tilde{\mu}, m_1, m_q^{-i1}] - \\ & \frac{1}{8} g_1 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} LF_{2,1,1,0} [\tilde{\mu}, m_2, m_q^{-i1}] - \frac{1}{4} g_1 m_2 \tilde{\mu} g_2^2 y_d^{i1i2} LF_{3,1,1,-1} [\tilde{\mu}, m_2, m_q^{-i1}] + \\ & \left. \frac{5}{24} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} LF_{2,1,1,0} [\tilde{\mu}, m_q^p, m_u^r] + \frac{1}{24} g_1 \tilde{\mu} y_d^{pi2} y_u^{i1r} \overline{a_u^{pr}} LF_{3,1,1,-1} [\tilde{\mu}, m_q^p, m_u^r] \right) \end{aligned}$$